Summer Eunhyung Ann

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Skills

Research: Multi-agent optimization, LLM evaluation, experimental design, causal inference, scientific AI, bioinformatics

Languages: Python, C++, SQL, R, MATLAB, Java, JavaScript/TypeScript, Swift

Frameworks & Tools: PyTorch, PyTorch Geometric, PyTorch Lightning, TensorFlow, Hugging Face, scikit-learn, NumPy, pandas, Optuna, NetworkX, LangChain, einops

Cloud & Systems: AWS (Bedrock, SageMaker, Lambda, EC2), Docker, persistent agent runtimes, distributed/async workflows, RAG, Node.js, Flask, Next.js

Education

University of Chicago *2024 – Present*
PhD, Computer Science
- Advisor: Chenhao Tan (since 2026). GPA: 3.98/4.00; full-ride scholarship.

University of Michigan – Ann Arbor *2021 – 2024*
BS, Computer Science & Biology, University Honors

Selected Publications

- **Summer Eunhyung Ann**, Haokun Liu, Chenhao Tan. "The Interaction Tax: When Communication Erases Diversity in Multi-Agent Teams." *ICML 2026 Workshop*. First author. arXiv; Code + data.
- **Summer Eunhyung Ann**. "On the Effect of Shared Memory on False-Belief Lock-In in Multi-Agent Systems." *ICLR 2027 submission*. First author. Paper; Code.
- Shiyang Lai, Arna Woemmel, Hongkai Mao, Junsol Kim, **Summer Eunhyung Ann**, James Evans. "Looking for Something Weird to Happen": How Humans Sustain AI Agent Novelty Amid Semantic Collapse. *CHI 2027 submission*. arXiv.

Research Grants

- **Anthropic Research Grant – $10,000 in API credits.** Awarded for "On the Effect of Shared Memory on False-Belief Lock-In in Multi-Agent Systems."
- **OpenAI API Credit Grant – $10,000 in API credits.** Awarded for research on safe cross-agent handoffs, verified human approval, and state-based evaluation of tool-using agents.

Research Experience

Researcher *Chicago, IL*
University of Chicago – Chenhao Tan Group *2026 – Present*
- Studied communication and search diversity across 11 verifier-scored optimization tasks under matched budgets in *The Interaction Tax*; showed that unrestricted full-solution exchange can collapse cross-model diversity, while independent proposals preserve coverage and often perform better.
- Design experiments and research infrastructure for multi-agent coordination, communication, interaction effects, and evaluation; investigate false-belief propagation through shared agent memory.
- Build scientific-agent infrastructure through Agent4Science and Flamebird, a persistent runtime for reproducible long-horizon research workflows.
- Analyze scientific discussion quality across 39K+ comments using causal/observational methods and study user influence on behavior in AI-only online social systems.

Researcher *Chicago, IL*
University of Chicago – Data Commons / Grossman Lab *2024 – Present*
- Work on AI-driven immune receptor modeling and large-scale biomedical data pipelines; build reproducible curation and quality-control workflows for AI-ready immune receptor datasets.

Research Collaboration – ProtoECGNet / EchoNext *2025*
Collaboration with Beaulieu-Jones Lab, University of Chicago
- Adapted prototype-based ECG models to the EchoNext cardiac dataset for interpretable structural heart disease classification, including embedding analysis and prototype-drift monitoring.

Contributor – Immune TCR Dataset Curation *2024 – 2025*
CZ Biohub Chicago
- Curated paired TCR alpha/beta sequence datasets from public sources for model-ready immunology data and downstream protein-language-model research.

Undergraduate Research Assistant *Ann Arbor, MI*
Michigan Medicine – Boyle Lab, University of Michigan *2022 – 2024*
- Applied SEI to STR genomics for disease classification and regulatory-activity prediction across 21,907 chromatin profiles.

Selected Technical Projects

Multithreaded Stock Trading System *C++20*

- Built a matching-engine simulation with multithreaded traders and order books; benchmarked 100K+ orders/second with near-linear scaling to 8 threads.

LLM Probing for Code Semantics vs. Syntax

- Constructed AST-preserving vs. semantics-altering probing datasets and hybrid subspace probes for code-model representation analysis.

TCR-GraphDiff & Biomedical Domain Adaptation

- Built graph-neural-network pipelines for TCR sequencing analysis and curated immune-receptor datasets for downstream modeling.
- Compared BERT and PubMedBERT representations for cancer classification with stratified/nested evaluation and calibration analysis.

DevSecCode – Deva *Jan 2025 – Present*

- Architect and build Deva, an AI-native secure coding platform combining rule-based scanning, AST-aware analysis, local reachability analysis, dependency/dataflow enrichment, calibrated confidence metadata, and LLM-assisted remediation.
- Developed editor-centered workflows preserving context through validation, remediation, control mapping, and evidence generation.

DevSecCode – Pasu

- Governance and control layer for automation and AI workflows, combining context-aware policy evaluation, configured approvals, scoped authorization, and retained execution evidence.
- Reproducible simulation and evaluation workflows with deterministic replay, restart/recovery checks, and cryptographically verifiable evidence bundles for governed actions.

Technical Experience

AI/ML Solutions Architect Intern *Seoul, South Korea*
Amazon Web Services (AWS) *May – Sep 2024*

- Designed and implemented ZON, an AWS Bedrock scheduling assistant for multi-step calendar reasoning and constraint satisfaction across time zones.
- Compared prompt engineering with function/tool calling, reducing invalid-timeslot and missing-constraint failures by approximately 30%; built structured logging, telemetry, and user-feedback loops to reduce hallucinated events.
- Deployed an EC2/OAuth 2.0 architecture tested at 10K+ concurrent sessions with sub-200 ms median orchestration latency in load tests.

LLM / Data Engineer *Remote*
Neuromuse.ai *May – September 2024*

- Integrated instruction-tuned LLMs into a natural-language-to-music stack with modular prompting and post-processing for style, tempo, and instrumentation control.
- Built evaluation pipelines combining semantic/lexical metrics with human preference studies, improving on-brief generation quality from 72% to 88%.
- Implemented text-audio dataset curation/deduplication and retrieval-augmented workflows for creator-specific conditioning without retraining the base model.

Co-CTO *Remote*
Concord Systems Corp. *June – August 2025*

- Built real-time MPC transaction-coordination tooling that increased internal benchmark throughput from approximately 50 to 200+ signed transactions/second.
- Integrated Fireblocks MPC approval flows, reducing end-to-end approval latency by approximately 40% and failed-signature incidents by 25%.

Teaching Experience

- University of Chicago: Natural Language Processing (Summer 2026); Foundations of Computational Data Analysis (Spring 2026); Machine Learning in Biology and Medicine (Winter 2026).
- University of Michigan: Information Visualization (SI 649 / EECS 548). Guided students in Python-based ML/data pipelines, visualization, and technical problem solving.